Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/02/2006
PI Name:	McDonald, Jay M.D.		
Project Title:	Molecular Mechanisms of Osteoblastogenesis Inhibition by Modeled Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Zip Code:	35294-0007	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	11/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to FSB as of FY2006, per program changes at that time, per J. Cohen/HQ and JSC/J. Dardano and previous information fr D. Risin/JSC in Mar 2006 (jvp 6/2009) NOTE: Funding reinstated to 11/30/2007, per J. Cohen (HQ, 8/2006)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):			
Grant/Contract No.:	NNJ04HB27G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Bone loss and increase fractures are serious consequences of spaceflight. Understanding the mechanisms responsible for hypogravity-induced bone loss is critical for developing effective countermeasures. One of the major factors contributing to this condition is a decrease in the ability of osteoblasts bone forming cells to form healthy well-mineralized bone. Our work is focused on examining the effects of hypogravity on the ability of the human mesenchymal stem cells (the cells which gives rise to osteoblasts) to become osteoblasts. We have found that hypogravity affects these cells and forces them to become fat cells instead of bone forming cells. This is consistent with true space flights where bone loss is accompanied with increased fat cells in the bone marrow. We are presently in process of testing different therapies to countermeasure this problem which can help in preventing bone loss in astronauts as well as elderly people here on earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our goal is to eventually discover a new therapy to prevent and reverse bone loss.
Task Progress:	Bone loss during prolonged space flight is a serious complication which must be addressed before extended space flight can be successful. Bones turnover at approximately the rate of 10% a year. An accelerated rate of bone loss over bone formation will result in an overall decrease in skeletal mass and a subsequent increase risk of fracture. The complex integration of the signals between the cells that build bones, osteoblasts, and the cells that destroy bones, osteoclasts, in the marrow is the focus of considerable attention by the scientific community. The successful development by NASA of the Rotary-Cell Culture System allows us to utilize our understanding of bone biology and the pathogenesis of bone disease to isolate various factors in laboratories on earth that affect bone integrity in microgravity during long-term space flight. Our research laboratory focuses on both cell types, osteoblasts and osteoclasts. We are investigating the signals whereby, the cells are produced (differentiate), how these cells function (activity), and how the cells die (apoptosis). It is the integration of these processes that allows us to understand bone turnover in humans. The effects of microgravity in human studies have shown initially what we thought to be predominantly a decrease in bone formation, but subsequent studies have indicated an increase in bone breakdown as well. The primary focus of our studies has been to determine the effects of simulated microgravity on osteoblastogenesis at the intracellular signaling level. We have utilized human mesenchymal stem cells which when treated appropriately develop into bone producing osteoblasts. If their development is altered, they may turn into fat cells, which occupy approximately 50% of the human bone marrow. There is also evidence from animal studies in space and from bed rest studies in humans on earth that the marrow fat cells increase under these conditions. Our studies to date have demonstrated that osteoblast precursors (human mesenchymal stem cells) in microgravity, instead of developing into osteoblasts developed into fat cells. The signaling process involved a decrease in the expression of collagen 1 (the main matrix protein in bone) and an increase in cell-binding integrins and subsequent three pronged signaling deficiencies: one that involves a series of protein kinases that result in decreased osteoblastogenesis; a second path that decreases the phosphorylation of cofilin, which results in decreased stress fiber formation and altered shape of the cells; and third an enhancement of nitric oxide signaling through actuation of nitric oxide synthase which will increase adipogenesis. Having demonstrated, this we can now employ our knowledge of osteoblastogenesis at the molecular level to reverse the effects of simulated microgravity. These agents include the calcineurin inhibitor cyclosporine, which may be an important target for reversing the effects of simulated microgravity. Other potentially effective countermeasure agents include BMP, statins, PTHRP, and IGF-1 to name a few. We will now also be able to test various agents that affect reactive oxygen species and protein phosphorylation. Although we would like to test these in a true microgravity environment on the international space station, this is not very possible at this point given the state of the program. Rather we will focus additional efforts on developing the unique system to study osteoclastogenesis (the cells that absorb bone) in simulated microgravity. We have preformed initial studies along this line and the system that we have developed using the rotary cell culture system will be ideal for studies, that very much parallel the studies on osteoblastogenesis. An ideal agent will inhibit osteoclastogenesis and stimulate osteoblastogenesis. The most likely candidates to date, low doses of the calcineurin inhibitors, cyclosporine and FR506, which inhibit NFAT activity, stimulating osteoblastogenesis, and inhibiting osteoclastogenesis.
Bibliography Type:	Description: (Last Updated: 02/19/2008)
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